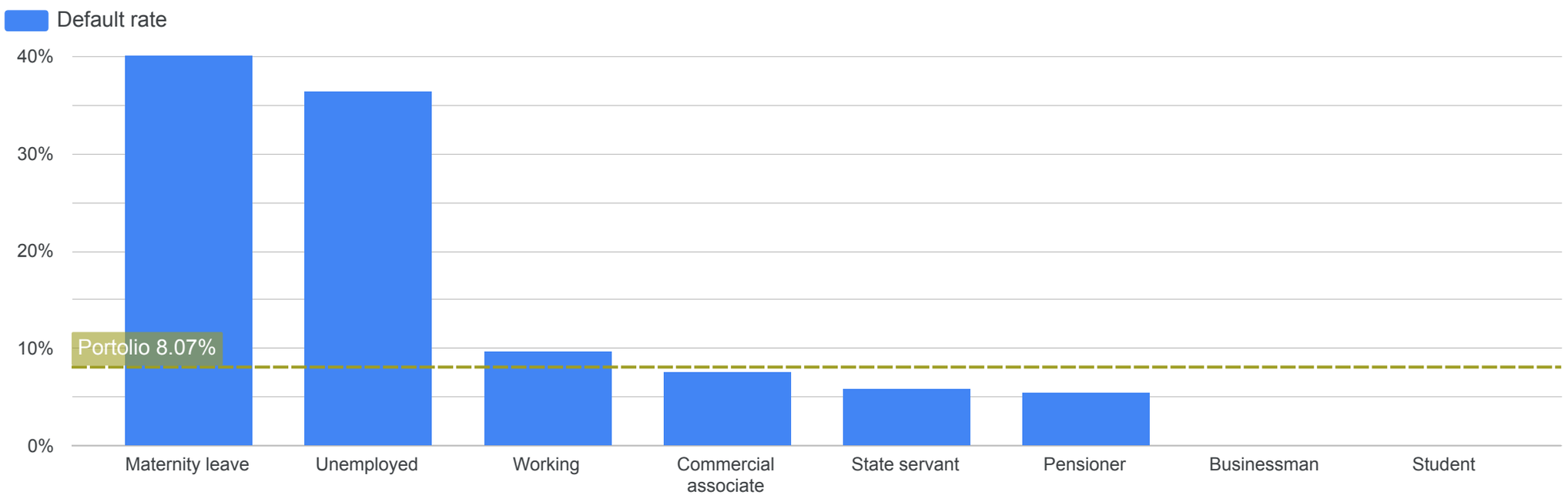
Executive Summary

Home Credit portfolio (307,511 applicants; model metrics on held-out test set)

Applicants	Default rat…	Test AUC (…	Test Gini (s…
307511	8,07 %	0,77	0,55

Default rate by income type



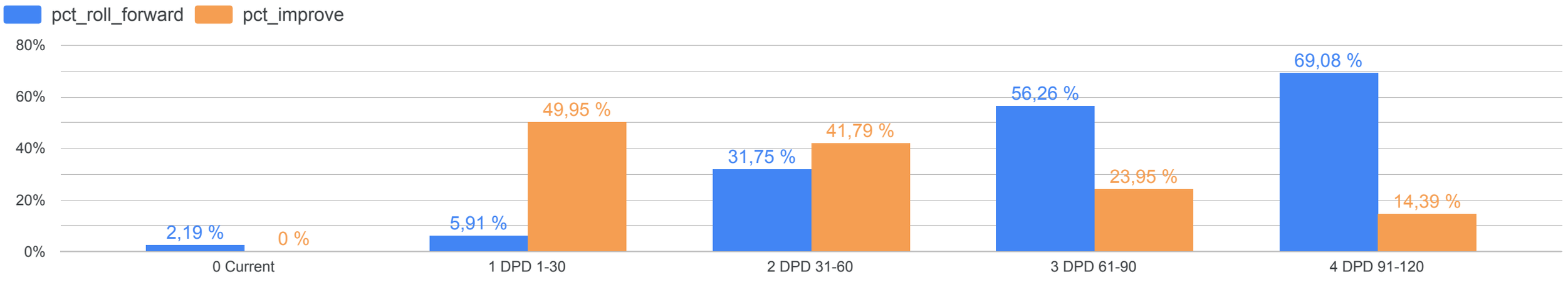
Delinquency Dynamics

Month-over-month status of Credit Bureau credits · all credit types · values rounded

Cumulative 90+ DPD rate by cohort

	To state / probability						
From state	0 Current	5 DPD 120+	4 DPD 91-120	1 DPD 1-30	3 DPD 61-90	6 Closed	2 DPD 31-60
0 Current	93%	0%	0%	2%	0%	5%	0%
1 DPD 1-30	50%	0%	0%	36%	0%	9%	6%
2 DPD 31-60	26%	0%	1%	15%	31%	8%	19%
3 DPD 61-90	13%	1%	55%	6%	12%	8%	5%
4 DPD 91-120	7%	69%	8%	3%	2%	8%	2%
5 DPD 120+	2%	94%	0%	0%	0%	3%	0%

Roll forward vs. cure by delinquency bucket

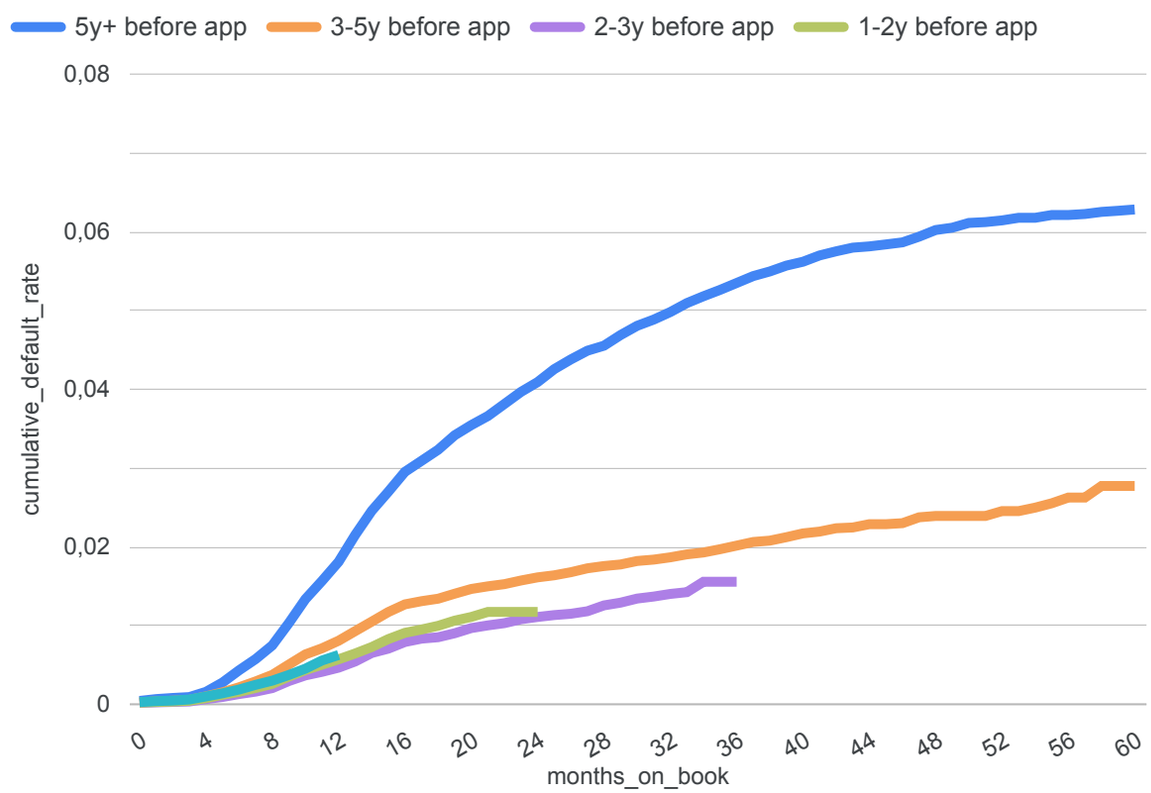


Key takeaways:
1) Only 5.9% of DPD 1-30 credits roll forward and 49.9% cure.
2) From DPD 31-60 the roll-forward rate jumps to 31.8%, then 56.3% and 69.1%.
3) DPD 120+ is almost absorbing (94% stay).

Vintage Cohorts

Cumulative 90+ DPD rate by months on book

Cumulative 90+ DPD rate by cohort



90+ DPD rate at 12 / 24 / 36 months

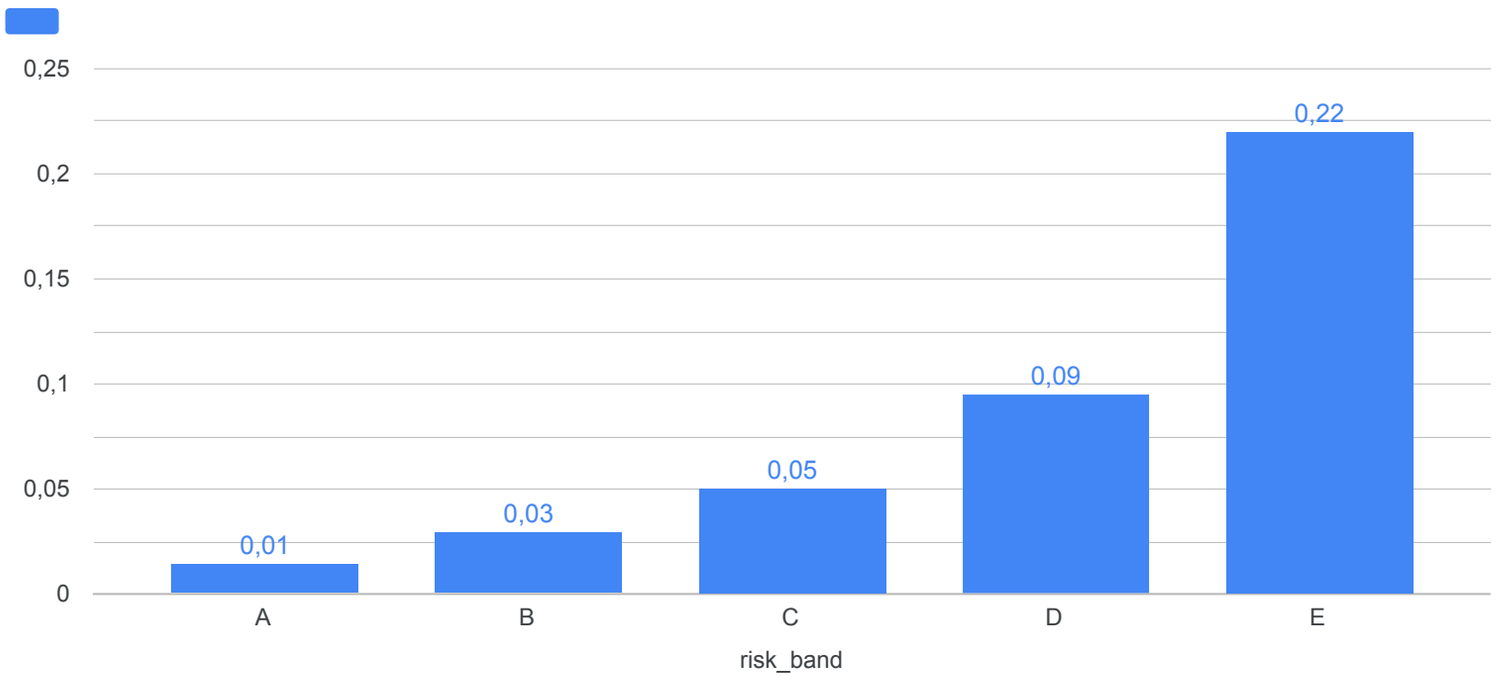
	months_on_book / cumulative_default_rate		
vintage_cohort	12	24	36
5y+ before app	1,8 %	4,08 %	5,34 %
3-5y before app	0,79 %	1,6 %	2 %
2-3y before app	0,45 %	1,1 %	1,54 %
1-2y before app	0,56 %	1,16 %	-
0-1y before app	0,61 %	-	-

Key takeaways:
1) The oldest cohort (credits opened 5+ years before the application) deteriorates fastest: 4.08% reach 90+ DPD by month 24 and 5.34% by month 36.
2) Cohorts opened 1-5 years before the application stay between 1.10% and 1.60% at month 24.
3) Cohorts are relative to the application date (the data has no calendar dates), so the gap reflects differences between credit generations, not a specific period. Portfolio reviews should track vintage curves to detect deterioration early.

Risk Segmentation

Applicants grouped into five risk bands (score quintiles defined on the validation set) · observed default rate on the held-out test set

Observed default rate by risk band (A = lowest risk)



Risk bands on the held-out test s…

	risk_band	n_clients	Observed…
1.	A	12.606	1,36 %
2.	B	12.278	2,88 %
3.	C	12.404	4,97 %
4.	D	11.930	9,45 %
5.	E	12.285	21,95 %

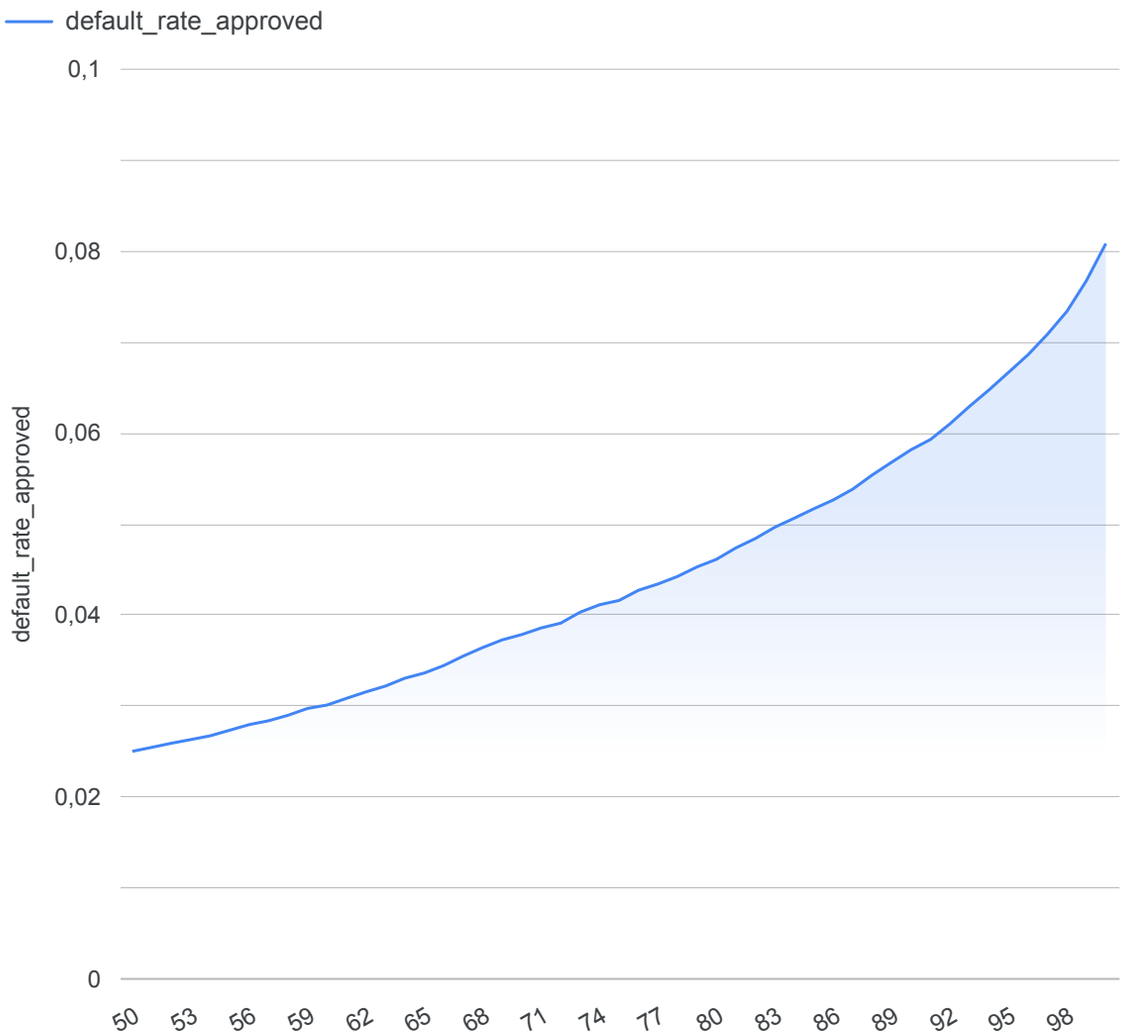
1 - 5 / 5

Key takeaways:
1) The model separates risk clearly: the default rate rises monotonically from 1.36% in band A to 21.95% in band E, a 16x difference.
2) Band E alone (20% of applicants) holds about 54% of all defaults in the test set.
3) Bands A-B (1.4%-2.9%) are candidates for faster or automated approval; band E (2.7x the portfolio average) should go to manual review or be declined.

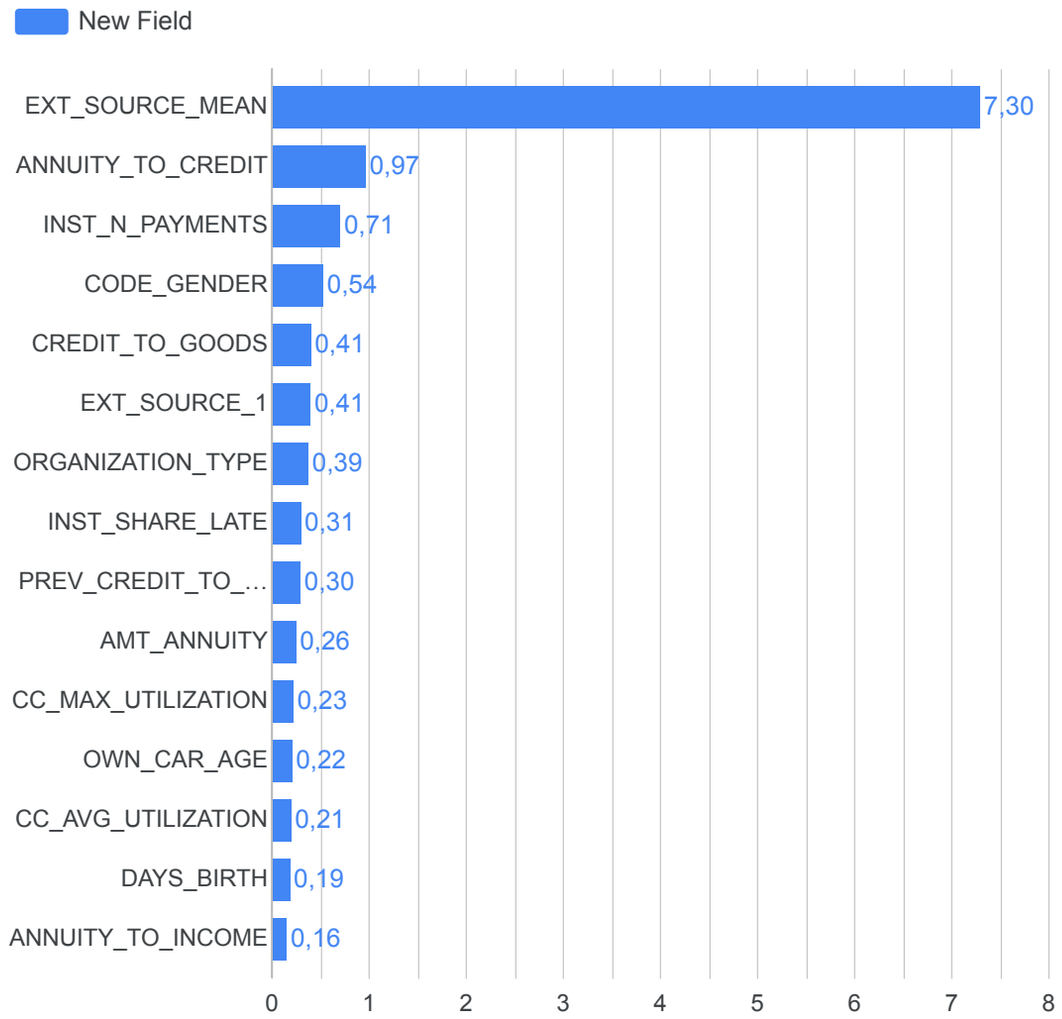
Approval Strategy & Model Drivers

Applicants ranked by predicted PD and approved from lowest risk upward · held-out test set (61,503 applicants) · simulated policy, not observed outcomes

Default rate of the approved book by approval rate



Top 15 drivers of the PD model (permutation importance)



Key takeaways:
1) Approving the 80% lowest-risk applicants lowers the default rate of the approved book from 8.07% to 4.60% and avoids 54% of defaults.
2) The curve steepens after ~85% approval: the last applicants approved add disproportionate risk.
3) External credit scores (EXT_SOURCE_MEAN) dominate the model; repayment behaviour (installment counts, late-payment share) and loan structure (annuity-to-credit, credit-to-goods) follow.